

Nhan Tran

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EDUCATION

Iowa State University

Master of Science in Computer Science (GPA 3.8 / 4.0)

Bachelor of Science in Computer Science (GPA 3.7 / 4.0)

Ames, IA

December 2025

May 2023

PROGRAMMING SKILLS

- **Languages:** C/C++ • C# • Java • Python • SQL • JavaScript
- **Softwares:** VS Code • Godot • Android Studio • Github • Git • Jira • AWS
- **Libraries:** Emscripten • WASM • Pytorch • Spring Boot

WORK EXPERIENCE

Principal Financial

Software Developer Intern

Ames, IA

June 2022 – August 2023

- Worked in an Agile environment among a team of software engineers.
- Developed and tested frontend and backend of various websites using Java and Spring Boot.
- Processed and visualized financial data using Python and SQL.
- Set up automatic gather, process, and visualize Energy Star certified housing data with AWS Athena, S3, DynamoDB.

Promax Unlimited

Software Developer Intern

Bettendorf, IA

January 2020 – June 2020

- Developed an authentication system for company's app using JavaScript and React Native.

PROJECTS

Procedural Generation Tile Constraints

April 2026

- Demo: pgtc.tranhqnhan.org
- Generated large textures, images from samples with locally self-similar structures.
- Implemented Model Synthesis / WFC on a 2D grid for procedurally generating textures and images from example.
- Multithreaded application to render the image and computing constraints in parallel.
- Developed with C++, Raylib, Emscripten, WASM, Multithreading.

Quadtree A* Search

March 2026

- Demo: qtas.tranhqnhan.org
- Improved speed of A* path planning on large grids by subdivide the grid with quadtree.
- Implemented quadtree algorithm with constant time neighbor finding to generate region graph.
- Developed with C++, Raylib, Emscripten, WASM.

Multiprocessing Task Scheduler with Ant Colony Optimization

November 2025

- Implemented Ant Colony Optimization algorithm for task scheduling in heterogeneous multiprocessing environment.
- Project for Principles of Operating System class.
- Developed with C and visualized with Raylib.

Iterated Dijkstra Propagation for Robotics Path Planning

November 2025

- Implemented Iterated Dijkstra Propagation for multiobjective path planning.
- Created a multicost resource pool architecture to manage memory efficiently.
- Developed an edge retrieval optimization that speed up the algorithm by seven percent.
- Developed using C++ and tested on the Husarion 2R robot with ROS platform.

AWARDS AND CERTIFICATIONS

Python Business Professionals of America Award

May 2022

Second place in Python programming competition hosted by Business Professionals of America.

Java IT Specialist

May 2022

Certified by Pearson Credly at Business Professionals of America event.

Python IT Specialist

May 2022

Certified by Pearson Credly at Business Professionals of America event.

Iowa Governor Scholar

May 2020

Awarded to the two highest academically performing seniors.

Hawkeye Programming Challenge

April 2019

Second place in programming competition hosted by University of Iowa